

Use of importance measures in risk-informed regulatory applications

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The use of importance measures to analyze PRA results is discussed. Commonly used importance measures are defined. Some issues that have been identified as potentially limiting their usefulness are addressed, namely: there is no simple relationship between importance measures evaluated at the single component level and those evaluated at the level of a group of components, and, as a result, some of the commonly used importance measures are not realistic measures of the sensitivity of the overall risk to parameter value changes; and, importance measures do not typically take into account parameter uncertainties which raises the question of the robustness of conclusions drawn from importance analyses. The issues are explored in the context of both ranking and categorization of structures, systems, and components (SSCs) with respect to risk-significance and safety-significance for use in risk-informed regulatory analyses. Published by Elsevier Science Limited.

1 INTRODUCTION

For several of the proposed applications of the risk-informed regulatory process, one of the principal activities is expected to be either the ranking or the categorization of structures, systems and components (SSCs) with respect to their risk-significance, or with respect to their safety-significance. A distinction is made between ranking and categorization. The purpose of ranking is generally to arrange items in order of increasing or decreasing importance. The purpose of categorization, on the other hand, is to allocate these items into two or more groups, according to some pre-set guidelines or criteria. Another distinction is made between risk significance and safety significance. Depending on the application, it may be appropriate to categorize or rank SSCs with respect to risk-significance, or with respect to safety-significance. In the next section, we propose definitions of risk-significance and safety-significance as applied to SSCs.

Various techniques, based on the use of importance measures of some kind, have been proposed for ranking and categorization of SSCs modeled in a PRA. For example, in the PSA Applications Guide¹ the Fussell–Vesely Importance, Risk Reduction Worth, and Risk Achievement Worth were specifically identified as appropriate measures to use. In his paper in the proceedings of PSA 96, Vesely raised a number of issues related to the use of importance measures in risk-informed regulation.² In this paper, the focus is on exploring more fully two of these issues, namely (a) the risk rankings apply only to individual contributions and not to combinations or sets of contributors, and (b) the risk rankings are not necessarily related to the risk changes which result from credible changes to contributor probabilities. Both these issues raise a concern about the validity of using component importance measures at all. In addition, we address the concern about the ranking being affected by uncertainties in the estimates of the probabilities of the basic events of the PRA model. First however, we define the Importance Measures, and introduce a Generalized Importance measure which addresses to some extent the second of the two issues on a single SSC level.

Finally, the relationship between the criteria for screening or ranking SSCs based on Importance Measures and the acceptance guidelines proposed for risk changes in the

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draft Regulatory Guide on the use of probabilistic risk assessment in risk-informed decisions on plant-specific changes to the current licensing basis³ is discussed.

2 RISK SIGNIFICANCE AND SAFETY SIGNIFICANCE

In this paper, risk significance and safety significance are regarded as complementary ways of identifying the role of SSCs in determining the risk from operation of the plant. Risk significance is perhaps easier to define in an operational sense than is safety significance. For example, an individual SSC can be identified as being risk-significant if it can be demonstrated that its failure or unavailability contributes significantly to measures of risk, such as, core damage frequency (CDF) and/or large early release frequency (LERF). The Fussell-Vesely importance measure, discussed later, is often used as a measure of risk-importance.

Safety-significance, on the other hand, is more difficult to define. Conceptually, safety-significance can be thought of as being related to the role the SSC plays in the prevention of the occurrence of the undesired end state. One way to capture the most safety-significant SSCs might be to identify those SSCs represented by the complementary cutset elements associated with the risk-significant SSCs. There is a weakness in this approach however, in that the reason why an SSC is not risk-significant, and may not even show up in the model solution, may be that the complementary SSCs are very reliable. In the extreme case, if there is only one compensating event (i.e. a two-element cutset), it would be hard to argue that the second event is not safety-significant, particularly if the failure probability of the first is relatively high. Following on from this, it could be argued that all the SSCs considered when constructing the PRA model (including those that do not necessarily appear in the final quantified model, either because they have been screened initially, assumed to be inherently reliable, or have been truncated from the solution of the model) have the potential to be safety-significant, since they play a role in preventing core damage.

In general, there are two types of applications in risk-informed regulation. The first results in increased attention being paid to the high significance group, and thus can be argued to result in a reduction in risk. An example of such an application might be the selection of SSCs for goal setting and monitoring as part of the implementation of the Maintenance Rule.⁴ (While, as pointed out by one of the reviewers of this paper, it can be argued that the increased attention in this case is not primarily to decrease risk, but to control the increase in risk from component degradation, this application does provide an improvement in the future risk over what would be the case were the rule not to be implemented.) In these cases, the categorization according to risk significance is more appropriate than the categorization with respect to safety significance since the SSCs that would

contribute most to a chosen measure of risk would be highlighted for increased attention. The risk ranking in this type of application can be based on the current best estimate risk model.

The second type of application is associated with the relaxation of requirements associated with the low significance group. Examples of such applications include the extension of test intervals for certain pumps and valves, the reduction of inspection frequencies for certain piping segments, or the grading of QA for SSCs. In these applications, caution should be applied when using risk significance to categorize SSCs. SSCs that might be thought of as low risk-significant as individual elements, may as a group, result in the loss of the redundancy or diversity of a set of equipment required for accident mitigation, to the point where the calculated risk increase would be significant. As an example, consider an Auxiliary Feedwater system with three pump trains. Because of the redundancy, the components in each individual train would not likely be risk-significant. However, because of the function played by the AFW system in the prevention of core damage, the components in all pump trains as a group could be safety-significant (as discussed in the next paragraph). Since a relaxation of requirements has the potential for resulting in an increase in risk, it is important to address the potential for this increase. Thus this would involve both the current best-estimate model, and a re-evaluation of that model with the projected change to the SSCs in the low significance group incorporated.

The identification and categorization of the safety-significant SSCs is not straightforward. However, one way to address this issue is to focus on the purpose behind the categorization. In the draft Regulatory Guide on Risk-Informed Regulation, DG-1061,³ a set of principles that must be addressed in an application for a change to the current licensing basis of a nuclear power plant is proposed. To be consistent with these general principles of risk-informed regulation, the impact of the change on overall risk measures should be related back to the acceptance guidelines for those risk measures. It is the proposed change that governs or dictates the categorization, and it is the potential impact of the change on the SSCs and on the measures of risk which ultimately determines which of the SSCs must be regarded as safety-significant. It is the group of SSCs for which changes can be made that satisfy these criteria that can be regarded as low safety-significant in the context of the application. Thus the real question that is being asked in the case of categorization is, for which group of SSCs can the change be made such that it meets the acceptance guidelines in terms of the change in risk, maintenance of the defense in depth philosophy, and the maintenance of safety margins. Thus an integrated decision-making process is essential in the categorization process.

To compare with the acceptance guidelines based on risk measures, it seems logical that the most appropriate way to address the categorization, particularly in the case of a

potential risk increase, is through a requantification of the risk measures. However, there are several applications, for example graded QA, for which there is no clear accepted approach to the evaluation of the impact of the change on SSC unavailability. It is for these applications in particular that the use of importance measures has been proposed as a means of establishing the membership of different categories.

3 IMPORTANCE MEASURES FOR INDIVIDUAL BASIC EVENTS

Originally, importance measures were defined for individual basic events of the plant logic model, as discussed below.

3.1 Commonly used measures

The three importance measures that have been most commonly used for ranking PRA basic events for use in regulatory applications^{5,6} are:

$$\text{Risk Achievement Worth } a_i = \frac{R_i^+}{R_0}$$

$$\text{Risk Reduction Worth } r_i = \frac{R_0}{R_i^-}$$

$$\text{Fussell - Vesely } FV_i = \frac{R_0 - R_i^-}{R_0} = 1 - \frac{R_i^-}{R_0}$$

where:

R_i^+ = overall model risk with the probability of basic event i set to 1;

R_i^- = overall model risk with the probability of basic event i set to 0;

R_0 = base (reference) case overall model risk.

The Risk Achievement Worth (RAW) yields the ratio of the model risk with the probability of basic event i set equal to 1 (the event has occurred or the equipment is failed) to the base case model risk. The RAW presents a measure of the 'worth' of the basic event in 'achieving' the present level of risk and indicates the importance of maintaining the current level of reliability for the basic event. As pointed out in Ref 7, the RAW of a component is not a very discriminating measure, and has to be interpreted very carefully. While it can be an appropriate measure for assessing a temporary change in which an SSC is to be made unavailable, if it is used in the context of assessing permanent changes, it is an extreme, bounding measure. However, it is commonly used as an intuitive measure of the margin provided by the component.

The Risk Reduction Worth (RRW) yields the ratio of the base case model risk to the risk with the probability of basic event i set equal to 0 (the event is impossible or the equipment is totally reliable) and represents the maximum decrease in risk for an improvement to the element

associated with the basic event. This measure is particularly useful for identifying improvements to the reliability of elements which can most reduce risk.

The Fussell-Vesely Importance is a measure of the fractional contribution of the basic event to the overall model risk when the basic event probability is changed from its base value to zero. As will be shown, the Risk Reduction Worth and the Fussell-Vesely Importance measures are related and the relative importance of basic events using both the RRW and FV measures is identical.

Some other measures which are used to assess the importance of PRA basic events include⁵:

$$\text{Birnbbaum Importance } I_B = R_i^+ - R_i^-$$

$$\text{Criticality Importance } I_{cr} = (R_i^+ - R_i^-) \frac{P_{i,0}}{R_0} = I_B \frac{P_{i,0}}{R_0}$$

where $P_{i,0}$ = probability of basic event i at its reference value.

The Birnbbaum importance (also called the reliability importance) is an interval risk importance measure which is completely dependent on the structure of the system model and is independent of the current probability of the basic event. The Criticality Importance measure is related to the Birnbbaum Importance. However, the basic event probability and current risk level are also considered in this importance measure.

These risk importance measures are relatively gross measures of the importance of a basic event and they generally suffer the following shortcomings: (1) they measure changes in risk and the importance of basic events only at the extrema (0,1) of the defined range of probability and (2) they do not consider the credible (uncertainty) range for the basic event probability. An alternative measure of importance which is based on credible changes to event probabilities is discussed in the next section.

3.2 A generalized risk importance measure

Schmidt *et al.*⁶ introduced the concept of a generalized importance measure which was defined as:

$$\frac{\Delta R_i}{R_0} = \frac{R_{i,n} - R_0}{R_0} = (R_i^+ - R_i^-) \left(\frac{P_{i,n} - P_{i,0}}{R_0} \right)$$

where:

$R_{i,n}$ = the model risk with a new value for the probability of basic event i ;

$P_{i,n}$ = the probability of basic event i at its new value.

Rearranging this equation slightly yields the general importance measure we will use in this discussion:

$$\frac{R_{i,n}}{R_0} = (R_i^+ - R_i^-) \left(\frac{P_{i,n} - P_{i,0}}{R_0} \right) + 1$$

Further rearrangement and the use of the importance

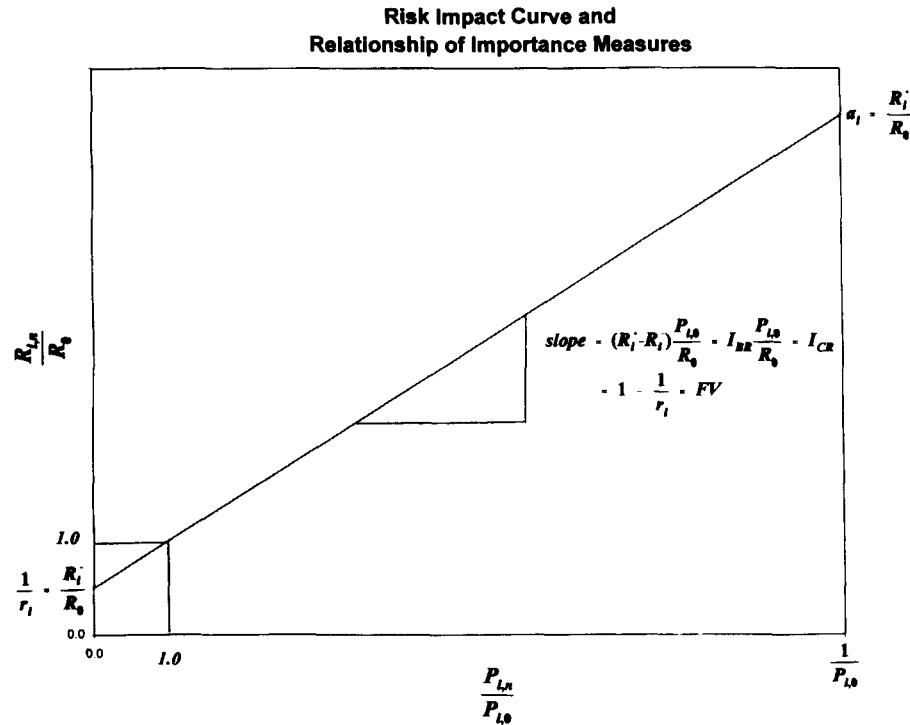


Fig. 1. Risk impact curve and relationship of importance measures.

measure definitions given above yields:

$$\frac{R_{i,n}}{R_0} = I_{cr} \left(\frac{P_{i,n}}{P_{i,0}} \right) + \frac{1}{r_i}$$

A similar relationship is derived in Ref 7. These importance measures are considered general since $P_{i,n}$ is allowed any valid probability and is not restricted to a value of 0 or 1 as is the case for a_i , r_i and FV_i . Since the generalized importance is defined for all values of $P_{i,n}$, a continuous relationship exists between $P_{i,n}$ and the generalized importance measure. Vesely *et al.*⁸ have termed this relationship a risk impact curve. (In this paper we will use the terminology—risk impact curve.) The relationship as expressed above is linear with a slope equal to the criticality importance and y-axis intercept equal to the inverse of the risk reduction worth ($1/r_i = R_i^-/R_0$). This general importance measure relationship has the additional property that when $P_{i,n}$ equals unity the general importance measure attains the value of the Risk Achievement Worth ($a_i = R_i^+/R_0$). Fig. 1 illustrates this relationship. As can be shown, and as indicated on Fig. 1 the FV and Criticality Importance measure are equal and are related to the Risk Reduction Worth (see Ref 6 for a clarification of the conditions under which the FV and criticality importance are identical):

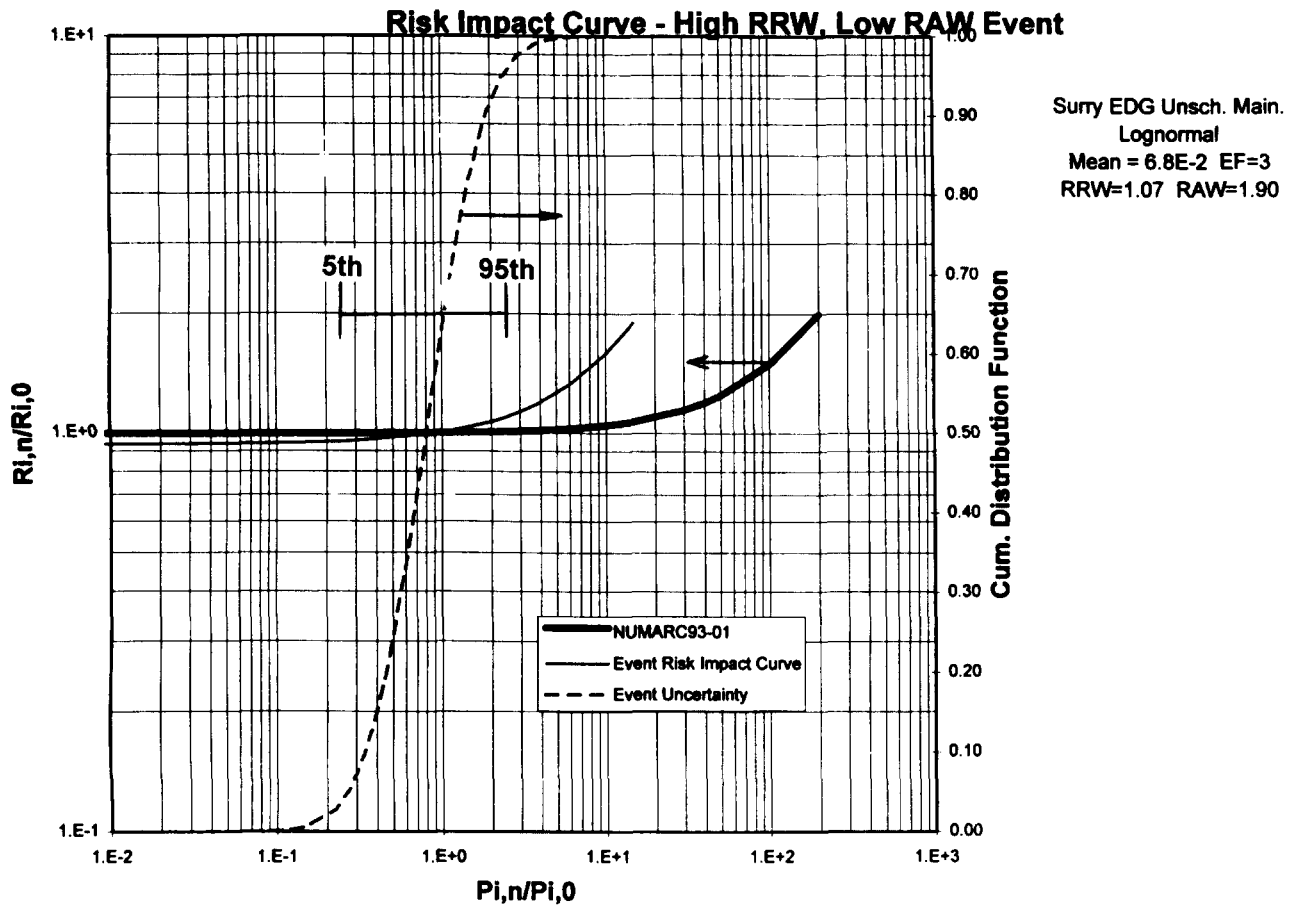
$$\text{slope} = I_{cr} = FV = 1 - \frac{1}{r_i}$$

Two illustrative examples of risk importance curves are

shown in Figs 2 and 3. These two curves are for events contained in the Surry Individual Plant Examination (IPE).⁹ Fig. 2 shows the risk importance curve for an event which has a relatively low Risk Achievement Worth and a relatively large Risk Reduction Worth (when compared with the criteria given in NUMARC 93-01¹⁰). Also shown on this figure is the risk importance curve (shown in bold) that envelopes those of the events which satisfy the NUMARC 93-01 criteria. The NUMARC 93-01 criterion for the RRW (and FV) importance is exceeded when the slope of the event risk importance curve is greater than that of the enveloping curve. (Note that in Figs 2 and 3, the risk importance curves have been plotted on a log-log plot to accommodate the range in probability values. A linear relationship, when plotted on a log-log scale has curvature. In these figures, the larger the slope of the linear relationship, the larger the curvature.) On both Figs 2 and 3, the uncertainties on the event probabilities have also been shown in the form of cumulative distribution functions.

Fig. 3 shows the risk importance curve for an event which has a large Risk Achievement Worth and a relatively small Risk Reduction Worth. The NUMARC 93-01 criterion⁸ for the RAW is exceeded when the risk ratio (y-axis value) at the right-hand side endpoint of the risk importance curve (at $P_{i,n} = 1$) exceeds a value of two.

Using the risk importance curves and the plausible change in basic event probability gives a different perspective on the risk importance of the basic events compared to the perspective gained by using only the



RRW and/or RAW. For the event shown in Fig. 3, little change in the risk occurs until the new failure rate is increased by more than an order of magnitude. Thus it could be concluded that changes which potentially may impact the probability of this basic event (such as changes to equipment maintenance, testing, Q/A, etc.) are of little concern unless they increase the basic event probability by an order of magnitude or more. On the other hand, for the event shown in Fig. 2 where there is a relatively greater slope in the risk importance curve, relatively small changes which either increase or decrease the basic event probability could have a significant risk impact. Use of this approach to characterizing importance can therefore give guidance on how significant the impact of a change has to be to have an impact on risk.

4 IMPORTANCE MEASURES FOR MULTIPLE BASIC EVENTS

A particular SSC may be represented in the logic model by several basic events. These different basic events can represent different modes of failure or unavailability of the SSC. A specific application may impact a subset of these modes. Therefore, to obtain a measure of the importance of an SSC

with respect to the particular application, all basic events representing the affected modes of the particular SSC should be considered as part of a group. Furthermore, many of the applications in risk-informed regulation are expected to have an impact on groups of SSCs, rather than on one individual SSC. Here again, the affected basic events have to be considered as part of a group. In a PRA model, it is the usual practice to estimate the unavailabilities of several SSCs on the basis of a single parameter, e.g., the same failure probability may be used for a large number of the MOVs in the plant. While the change being proposed may not impact all the MOVs, it would affect a subset, for example, those in a redundancy group. The appropriate way to model the impact of the change is to modify the relevant parameter for that subgroup. Thus, it is the acceptability of the parameter change for that subgroup that is in question. An additional complication that may have to be addressed for many applications is that the group of SSCs considered for the change may be inhomogeneous. For example, it may include MOVs, AOVs, and pumps. Thus, there may be a number of failure rates or probabilities that are affected. Since the overall increase in the risk indices due to the change is what has to be controlled, what is of interest is not the importance ranking or categorization of the basic events representing the affected modes of failure or unavailability of

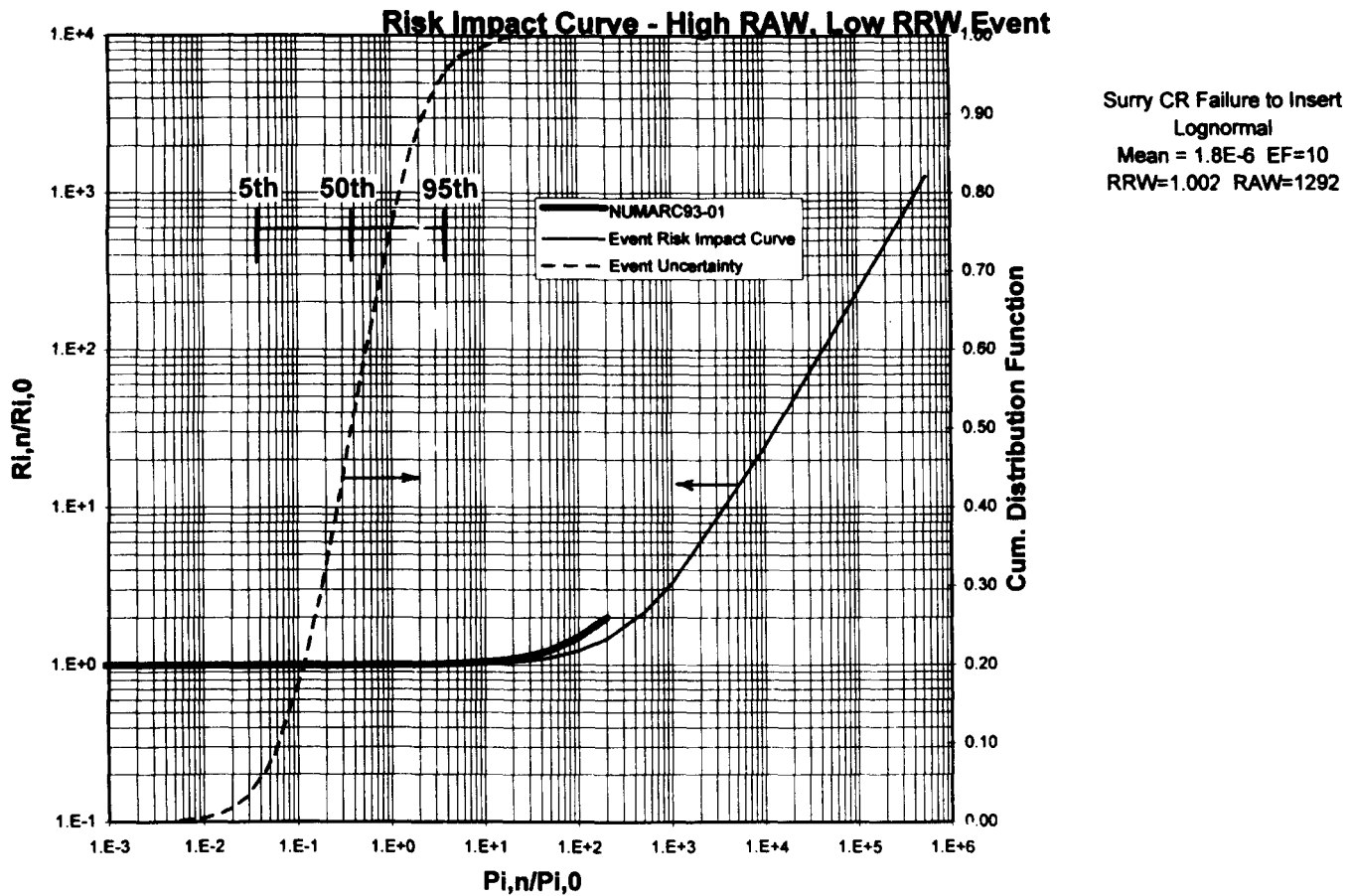


Fig. 3. Risk impact curve—High RAW, low RRW event.

these SSCs taken one at a time, but of the SSCs taken as a group. It is the integrated impact of the changes that is of concern.

As demonstrated in Appendix A, there is no simple relationship between the importance measures of the individual members of a group and the group taken as a whole. One question that needs to be answered therefore is whether it is possible to develop importance measures for the groups of basic events that are the subject of a change. There are really two parts to this question. The first is whether the importance measures have a useful physical interpretation, and the second is whether a meaningful group importance measure can be established.

basic events would be to evaluate the RAW by setting the probability of each member of the group to 1. That this is not an appropriate solution can be easily shown. Suppose the cutset equation is of the form:

$$R = AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH$$

Following the approach used in many of the standard PRA software codes, a simple substitution of probabilities for each of the events, but setting the probability for each member of the group C to 1 would result in a RAW of

$$RAW(C) = \frac{2AB + 2DE + 4F + GH}{AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH}$$

The significance of a group of basic events, if it is to be measured using the equivalent of RAW, FV and RRW, cannot be obtained directly from the individual basic event importances. Potential group importance measures are discussed below for those measures commonly used.

4.1 Risk achievement worth

One possible approach to evaluate the RAW of a group of

An interpretation of this equation is that, for components in series, the effective unavailability of the series group is n , where n is the number of components in series. Thus, the effect of this simple substitution is the generation of non-minimal cutsets, and the group RAW would therefore be larger than expected, with the magnitude of the difference being dependent on the number of components appearing in series in the logic model. This results because fault trees typically do not include success states. This also suggests

that the substitution of 1 for each of the components in a system is not a good way of evaluating system importance.

Another approach is to evaluate the RAW for each of the basic events of the group and add them. The RAW(C_1) for a single basic event taken in the usual way is given by

$$\text{RAW}(C_1) = \frac{AB + ABC_3 + DE(C_2 + C_4) + F(1 + C_3)(C_2 + C_4) + GH}{AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH}$$

That this has similar problems to the above formation can be seen by noting that the numerator of this sum would contain a contribution of 4GH.

If the cutset equation were to be evaluated as a probability equation with appropriate Boolean reduction, as outlined below, this problem would be avoided:

1. Take the cutset equation for the measure of interest (CDF, LERF).
2. Rename the basic events representing the SSCs in the group under investigation so that they all have the same identifier.
3. Boolean reduce the cutset equation.
4. Calculate the risk index for the new basic event with its value at unity, and calculate the RAW using the original, unmodified cutset equation of the base case evaluation (i.e. the denominator).

This gives an importance measure for the group. In this case the group is totally correlated in its effect as the probabilities of the individual members of the group are set to one. The difference between this approach and that taking the group event probabilities individually to one or zero can be seen in the following.

The substitution of C_1, C_2, C_3, C_4 by C and the re-reduction of the equation would give

$$R = ABC + DEC + FC + GH$$

and

$$\text{RAW}(C) = \frac{AB + DE + F + GH}{AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH}$$

It can easily be seen that this approach to evaluating a

group RAW cannot be expressed simply in terms of combinations of the RAW measures for the individual members of the group.

4.2 Birnbaum importance measure

The Birnbaum importance of an individual basic event is evaluated by $I_B = R_i^+ - R_i^-$. For a single basic event, the sensitivity of the risk measure to the probability of that event, p , can be parameterized as $R(p) = I_B p + R_i^-$. The value of the Birnbaum importance measure therefore is that it represents the sensitivity coefficient of the risk measure to

the probability of that basic event, and provides one way of looking at the defence in depth issue in a probabilistic sense.

That it is difficult to generate a meaningful group Birnbaum measure that can be used as a sensitivity measure

can be seen by considering that, when there are terms in the equation for R which involve multiples of p , this simple relationship breaks down.

It might be thought that using the procedures discussed above for the RAW, might produce meaningful measures. Then, for example, in evaluating a group importance using the same equation for R above, if a substitution of 1 were made for each member of the group C , the resulting 'group Birnbaum measure' would be

$$I_B(C) = 2AB + 2DE + 4F$$

Following the same substitution and remineralizing procedure as suggested for RAW would result in a 'group Birnbaum measure' of $AB + DE + F$. That neither of these is an appropriate sensitivity measure can be seen by comparing them with the sensitivity measure evaluated in a later section.

4.3 Fussell–Vesely importance

The Fussell–Vesely measure of importance for a single basic event basically represents the fraction of the risk measure to which the basic event contributes, i.e. it is the sum of the cutsets involving the basic event divided by the sum of all the cutsets. The Fussell–Vesely measure obtained by including all cutsets that contain one or more basic

events of the group is given by:

$$\text{FV}(C) = \frac{AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4)}{AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH}$$

This is a measure that assesses the contribution of the group in such a way that, any cutset that has a contribution from any one member of the group is included. Note, however, that this is not the same result that would be obtained by adding the individual Fussell–Vesely measures. Since this measure does not involve assessing changes, but is a simple ratio of contributors, this is an appropriate measure of group importance.

4.4 Risk reduction worth

The risk reduction importance of a single basic event is the

ratio of the CDF to the CDF calculated with the probability of the basic event set to 0. Substituting 0 for each member of the group to calculate the RRW is an appropriate way to calculate group importance since, in this case, there is no problem with non-minimal cutsets.

4.5 Generalized importance

Since the use of this measure is based on a linear sensitivity equation, it will run into the same problems as those discussed for the Birnbaum importance measure. This is explored more fully in the next section.

5 RELATIONSHIP OF IMPORTANCE MEASURES TO RISK CHANGES

An issue that has been noted by Vesely², among others, is that the importance measures are, for the most part, not directly related to the risk changes associated with the change being evaluated. That this is true for those importance measures which are based on taking parameter values or basic event probabilities to their extremes should be obvious. The one exception appears to be the generalized importance measure, and only then when the SSCs for which the change is being evaluated are unique, i.e., the associated basic events only occur once in each cutset in which they appear, and are independent from all other basic events.

We have not been able to identify any one sensitivity measure related to importance measures that can fill the role of characterizing the change in risk, particularly when a group of components is affected by the change. As a simple example, take a cutset equation and treat it as an algebraic equation, replacing each of the events in the group of interest by a common parameter, and without performing a Boolean reduction, differentiate the equation with respect to that parameter. For the equation $R = AB(C_1 + C_3) + DE(C_2 + C_4) + F(C_1 + C_3)(C_2 + C_4) + GH$, the algebraic equation would become $R = 2ABC + 2DEC + 4FC^2 + GH$, and

$$\frac{dR}{dC} = 2AB + 2DE + 8FC$$

This is a sensitivity parameter that is valid when the changes in the value for C are small, and the approximation

$$\Delta R = \frac{dR}{dC} \Delta C$$

is appropriate if the impact of the change on each member of the group is the same. This sensitivity parameter is, however, different from any of the importance measures derived in the previous sections. As the magnitude of the changes in C increases, higher order derivatives are needed to assess the change in R .

Thus, it can be concluded that the sensitivity of risk to a multi-component change cannot easily be related to component level importance measures.

6 THE ROLE OF IMPORTANCE ANALYSIS IN DETERMINING SAFETY-SIGNIFICANCE OR RISK SIGNIFICANCE FOR USE IN RISK-INFORMED REGULATION

Given the above discussion, it is reasonable to question the extent to which component level importance measures can provide insights to assess the risk-significance and safety-significance of groups of components, and whether other, more appropriate measures can be established.

6.1 Risk-significance

The group FV measure, where the numerator of the expression is obtained by summing the probabilities of all the cutsets which contain at least one basic event of the group appears to be a reasonable measure to assess the impact of the group of components before the change. Therefore, it is a reasonable measure of the risk-significance of the group. This should not be taken to imply that individual basic event importances are not of value. It is likely that, within the group of interest, because of different roles of individual SSCs, or because of asymmetries for example, some components, and specifically, some failure modes of components, may be more 'important' than others.

This group importance measure cannot, however, be related to the impact of the change since it is not a sensitivity parameter, and furthermore merely reflects the status quo. A relevant question to ask is, at what value of this group FV parameter can the group be considered to be significant. The current component and system level FV significance criteria, such as those given in Ref 11, are chosen to represent a small (0.5% (component) and 5% (system)) contribution to the total risk measure. The choice of criterion essentially determines how much of the risk is being controlled during the change.

It is important to note that setting a fixed FV measure to screen for significance means that, for plants with different base CDFs or LERFs, the contributions to these risk measures represented by the basic events are, in an absolute sense, different. Therefore, if we are interested in controlling the change in risk in an absolute sense, it does not make sense to have a universally fixed value of FV as a criterion for risk-significance, since the FV Importance measure is a relative measure (to the base risk).

6.2 Safety-significance

One approach that has been proposed as a means of identifying safety-significant SSCs is to use the RAW. The RAW of a basic event is essentially a measure of the impact of removing the particular basic event (i.e. setting its probability to one). The group RAW measure, evaluated using the substitution and reduction technique suggested previously, seems a reasonable approach to dealing with a group. However, it has to be recognized that RAW is a very extreme measure, and that there are very few SSCs, if any, for which

the impact of a proposed change is to render them totally ineffective. If RAW is used as a criterion for categorization, there is an important issue associated with the choice of an appropriate numerical value.

It is important to realize that, if the categorization is to be used in the context of risk-informed regulatory decisions based on the safety principles and acceptance guidelines in DG-1061³, that, using a single value of RAW as a universal criterion for establishing group membership can lead to inconsistent treatment of different plants. Consider the following example:

Plant X: $R = A \cdot B + C$, where $A = 1\text{E}-4$, $B = 1\text{E}-1$, $C = 1\text{E}-4$, $\text{RAW}(B) = 1.818$;
 Plant Z: $R = A \cdot B + C$, where $A = 1\text{E}-5$, $B = 1\text{E}-1$, $C = 1\text{E}-5$, $\text{RAW}(B) = 1.818$.

Now suppose that a universal screening criterion for safety significance is chosen using a RAW value of 2. In this case both plants would classify their respective component B as low safety significance. Suppose further that the change being considered had been assessed as increasing component B 's unavailability by a factor of two in both cases. In this case the increase in risk for plant X would become $1\text{E}-5$, whereas, for Plant Z it would become $1\text{E}-6$. Thus the same criterion would result in different impacts on risk depending on the risk profile of the plant, and, in particular, on the base risk value.

The question again, is how to set the criterion for categorization into low or high safety significance, using RAW, recognizing that, for a fixed value of RAW, the change in allowable risk would be different from plant to plant, that the acceptance criteria for the increase in risk applies to the whole group, and that the setting of the event unavailabilities to unity is generally an extreme, unrealistic way of modeling the impact of the proposed plant change.

One possible approach would be to accept the potentially unrealistic nature of RAW in the first instance, and to tie the RAW back to the allowable increase in risk (ΔR). If this were the case, the allowable RAW would be determined by $(R + \Delta R)/R$. Thus RAW would depend on the base R , and, given the acceptance criteria in DG-1061 for CDF as an example, could take any value above 1.01–1.1 depending on the base CDF. Following this approach, it is clear that it does not make much sense to define a universal criterion based on RAW. This value of RAW applies to the whole set of basic events. If it were considered useful to translate this to a component level criterion, it would be necessary to back out values of RAW at the individual basic event level that are consistent with this group value. This is clearly dependent on the nature of the group, and, in particular, when the elements of the group are inhomogeneous, this is not easily done. It points out however, that, for different applications, with different groups of SSCs, there cannot be a universal screening criterion at the basic event level.

Because of the extreme nature of the RAW measure, it is likely that the SSCs identified as safety-significant by this process would be a large set. At this stage, the simplest

course would appear to be to perform a set of well chosen sensitivity studies, based on realistic bounding estimates of the changes on event unavailabilities to identify which of the SSCs in the group might be dropped out of the category. Thus, by a process of iteration, it should be possible to arrive at a realistic identification of the non-safety-significant SSCs.

The above approach can be thought of as a top-down approach to categorization. Another approach would be to build up the non-safety-significant category using a bottom-up approach. In practical terms this could be achieved by ranking the SSC basic events by their individual RAW values, then starting from the bottom of the list, building up a set that satisfies the group RAW criteria developed from the acceptance guidelines and the plant specific CDF and LERF. A similar approach was suggested by the Boiling Water Reactor Owners' Group (BWROG) in their topical report describing categorization of MOVs in response to Generic Letter 89-10, for reviewing the categorization of the initially low ranked MOVs. For this group, they suggested a substitution of 1 for each of the MOVs, and using the increase in CDF to judge whether the MOVs could really be considered as low contributors. DG-1061 gives appropriate limits on the increases in CDF and LERF. A straight substitution of 1 for each of the MOV values results in a conservative estimate of the potential increase in CDF or LERF. Replacing all the SSC basic events by the same event name and re-minimalizing the cutset equation would be a less conservative approach. This bottom-up approach can be made fairly flexible. For example, it is not necessary to sequentially add in the SSCs from the bottom of the list; the list could be made up by picking and choosing from the low ranked SSCs. This would likely require several attempts to achieve the optimal set of SSCs.

7 DEMONSTRATION OF ROBUSTNESS OF CONCLUSIONS FROM IMPORTANCE ANALYSES

Importance analyses are only as credible as the logic model upon which they are based. Furthermore, whatever approach is used, importance measures on their own cannot be used without a demonstration that the results are robust with respect to a variety of uncertainties in the model. Since the only uncertainties that are typically treated formally in PRAs are the parameter uncertainties, the discussion below will focus on those uncertainties and their impact on importance measures.

7.1 Parameter uncertainties and their impact on importance measures

Consideration of the risk importance curve discussed earlier, in conjunction with the basic event uncertainty distribution provides insights into the robustness of the event importance to risk with respect to the uncertainty in the event probability. For example, for the event shown on

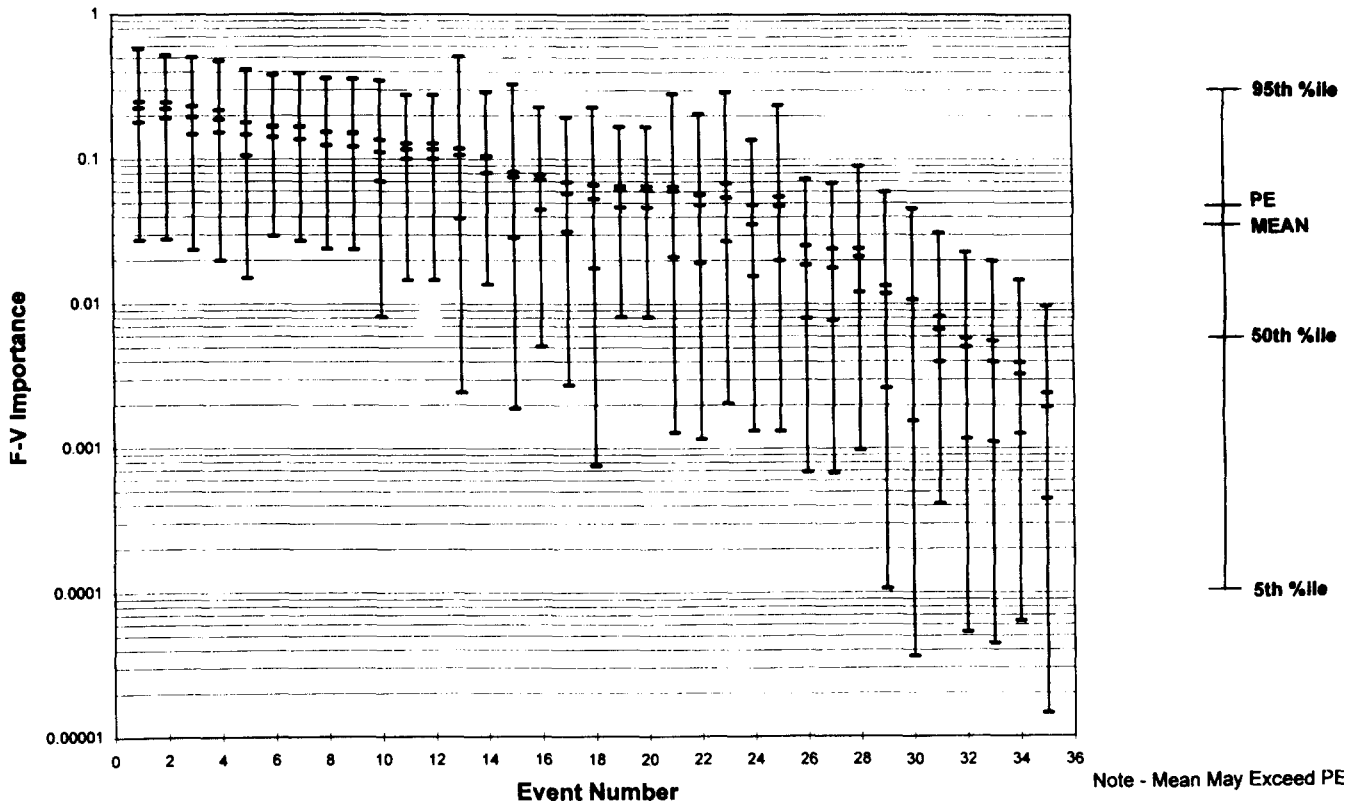


Fig. 4. Uncertainty in Fussell-Vesely importance.

Fig. 3 (an event with RAW equal to 1292) the increase in risk is less than 1% when evaluated at the 95th percentile in the event uncertainty distribution. This result indicates that the change in event importance when the parameter value is varied over the uncertainty range for the event probability is relatively low. By contrast, the very large value for the RAW would suggest a much greater importance than might be warranted. The RAW is measuring the risk significance if the event failure probability changes from its base value to unity, without considering how credible is the assumption that the event probability equals unity. On the other hand, for the event shown in Fig. 2 (high RRW, low RAW) the slope of the risk importance curve is sufficiently large that the risk increase at the 95th percentile in its uncertainty distribution is significant, representing an approximate 10% risk increase. Hence, the importance of this event is more sensitive to the uncertainty on the parameter value uncertainty characterization.

However, this only addresses the uncertainty associated with the event whose importance is being evaluated. The following discussion summarizes a methodology for assessing the uncertainties associated with risk importance measures, taking into account the uncertainty on all parameter values. It should be noted that the discussion in this section concerns the uncertainty on the importance measures themselves. There are other importance measures, called uncertainty importance measures which are intended to identify the SSCs whose uncertainties contribute most to the uncertainty on the risk measures (see for example,

Ref 11). However, these are not discussed here. This discussion will focus on the FV Importance measure. The FV Importance for an event X_i is the ratio of the frequency of system failure resulting from failure of event X_i to the total system failure frequency. In terms of the system minimal cutsets, the FV Importance is the frequency of the union of all the cutsets containing X_i divided by the frequency of the union of all the system model cutsets.

$$FV_i = \frac{F\left(\bigcup_{j=1}^m C_j\right)}{F\left(\bigcup_{k=1}^n C_k\right)}$$

where:

C_j are all cutsets containing Event X_i , and
 C_k are all the model cutsets.

Using the above equation, the uncertainty in the FV Importance is determined by propagating the basic event uncertainties throughout the PRA model using simple Monte Carlo (MC) or Latin Hypercube (LHC) simulation to determine the distribution on the FV Importance for each event.

Example results for a set of basic events from the NUREG-1150¹² Surry model are shown on Fig. 4. These events represent the top 25 events (ranked by point estimate FV Importance) and a number of other events with FV Importance decreasing down to below the NUMARC 93-01 criterion for the component FV Importance (0.5%).

Table 1. Summary of events from NUREG-1150 Surry Plant IRRAS model included in study

No.	Event name	FV-Importance	Description
1	NRAC-7HR	0.251	Non-recovery AC pwr w/in 7 h of LOSP/
2	OEP-DGN-FS-DG01	0.248	Diesel generator #1 fails to start
3	QS-SBO	0.233	Power conversion system: Stm gen integrity
4	REC-XHE-FO-DGHWB	0.219	OP fails to rec a DG FM HW fail in 6 h
5	REC-XHE-FO-DGEN	0.180	OP fails to recover a DG within 1 h
6	OEP-DGN-FS	0.169	Diesel generator fails to start
7	REC-XHE-FO-DGHS	0.169	OP fails to rec a DG FM HW fail in 3 h
8	RCP-LOCA-750-90M	0.155	750 GPM RCP SEAL LOCA at 90 min
9	NRAC-216MIN	0.152	Non-recovery AC pwr w/in 216 min of LOSP
10	NRAC-1HR	0.136	Non-recovery AC pwr w/in 1 h of LOSP
11	OEP-DGN-FS-DG02	0.129	Diesel generator #2 fails to start
12	OEP-DGN-FS-DG03	0.129	Diesel generator #3 fails to start
13	OEP-DGN-FR-6HDG1	0.118	DG 1 fails to run 6 h
14	BETA-3DG	0.104	Beta for CC failure of 3 DGs
15	BETA-2MOV	0.082	Beta for CC failure of 2 MOVs
16	REC-XHE-FO-SCOOL	0.078	OP fails to get RCP seal cool during SBO
17	AFW-XHE-FO-CST2	0.070	Failure of OP to XCONN unit 2 CST
18	SBO-PORV-DMD	0.066	PER valve RCS PORV demand PROB during SBO
19	BETA-2DG	0.065	Beta for CC failure of 2 DGs
20	NOTDG-CCF	0.065	Success of the third DG after CCF of 2
21	OEP-DGN-FR-6HDG2	0.065	DG 2 fails to run 6 h
22	OEP-DGN-FR-6HDG3	0.057	DG 3 fails to run 6 h
23	AFW-XHE-FO-UNIT2	0.055	OP fails to XCONN AFW transients
24	OEP-DGN-MA-DG01	0.049	Test and maintenance on diesel generator 1
25	R	0.048	Failure to manually scram the reactor
26	OEP-DGN-MA-DG02	0.025	Test and maintenance on diesel generator 2
27	OEP-DGN-MA-DG03	0.024	Test and maintenance on diesel generator 3
28	LPI-MOV-PG-1890C	0.021	LPI motor operated valve 1890C plugged
29	LPR-MOV-FT-1860A	0.013	LPR motor operated valve 1860A fails to open
30	HPI-XHE-FO-ALTS3	0.011	OP fails to rec CCF of HPI discharge MOV
31	HPI-CKV-FT-CV410	0.007	Check valve CV410 fails to open
32	HPI-XHE-FO-UN2S2	0.005	OP fails to XCONN HPI to U2 for S2-D1
33	AFW-MDP-FS	0.004	AFW MDP fails to start
34	AFW-XHE-FO-U1SBO	0.003	OP fails to XCONN AFW SBO at unit 1
35	LPR-XHE-FO-HOTLG	0.002	OP fails to align the system for hot leg RECI

Table 1 provides a description of these events and their point estimate FV importance. The results shown on Fig. 4 provide the 5th, 50th and 95th percentile values and the mean values from the distribution determined from Latin Hypercube simulation. Also shown are the calculated point estimate (PE) FV Importance values.

An important item to note from this figure is that the spread of the uncertainty distribution (5th to 95th percentile) is relatively broad—particularly at lower values for the FV Importance (approaching three orders of magnitude). The broad uncertainty distributions for these events can pose a challenge when ranking events based on risk importance since a significant overlap in the distributions for events indicates that there may be significant uncertainty in the relative importance and ranking for the events.

To obtain a better understanding of the significance of the uncertainties on the importance measures to risk ranking, several additional analyses were performed. It was recognized that the event FV uncertainty distributions are not independent. In fact, they may be highly correlated. Three principal sources of correlation can be identified. Firstly, the uncertainties in individual basic events may be correlated. For example the uncertainties in the probability of diesel

generator (DG) failure to start are correlated for DGs 1, 2 and 3 in the Surry model. Secondly, events which are basic events of the same minimal cutsets will have their FV Importance correlated. Thirdly, all events are correlated through the denominator in the FV Importance which contains the union of all cutsets.

In order to assess the extent of the correlation, the linear correlation coefficient between the FV Importance was calculated for all combinations (pairs) of events included in the calculation. Strong correlation was found to exist between a number of events. For example Events 2, 11 and 12 (Diesel Generators 1, 2 and 3 Failure to Start) have correlation coefficients exceeding 0.99. This strong correlation results from the fact that the uncertainties in the basic event uncertainty distributions are assumed to be correlated in the model and these events tend to appear together in many dominant cutsets. Event 1 (Failure to Recover AC Power Within Seven Hours of a Loss of Offsite Power) is more loosely correlated to Events 2, 11 and 12 (correlation coefficients between 0.4 and 0.5). In this case the basic event uncertainty distributions are not correlated; however these events occur together in many cutsets.

As a measure of the robustness of ranking two events

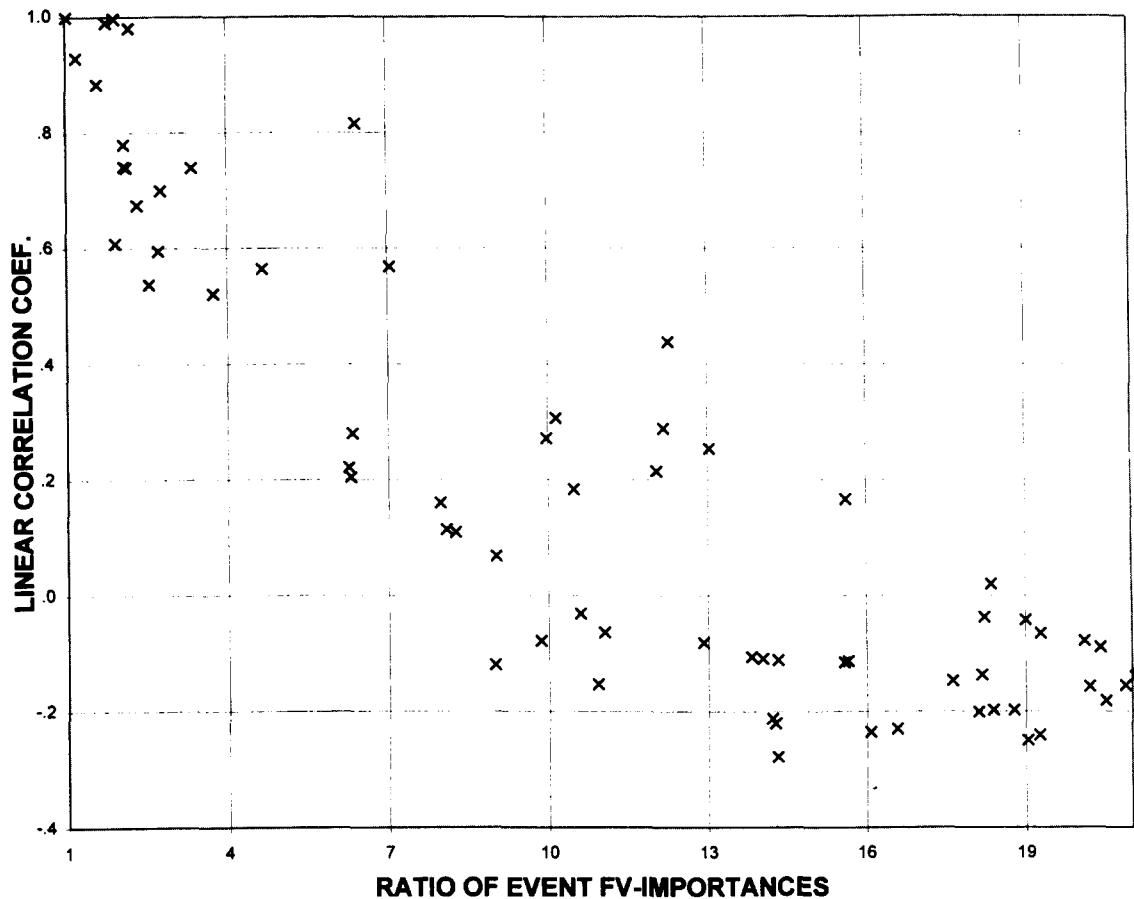


Fig. 5. Ratio of FV importances—High confidence ranking.

based on point estimate importance measures, calculations were performed to assess the probability that the FV Importance of one event exceeded the FV Importance of a second event. This calculation was performed by counting the number of observations from the Latin Hypercube simulations where the FV Importance for the first event exceeded the FV Importance of the second event. A score of 95% was selected as the test measure for robustness—that is, the FV Importance of the first event must exceed the FV Importance for the second event in 950 out of 1000 observations. For all cases where this test was met it was judged that the first event could be confidently assumed to have greater importance to risk than the second event.

Finally, to translate this information into a form that provides meaningful information, the ratio of the point estimate FV Importance for each event pair combination that passed the robustness test were plotted against the calculated correlation coefficient between these events. This plot is shown in Fig. 5. The information on this figure suggests that a difference of about one order of magnitude (ratio of FV Importance of ten) is necessary for a high degree of confidence that one event is more important than another unless significant correlation exists between the events (i.e. their uncertainties are associated and/or they consistently appear together in cutsets).

An alternative approach to ranking under uncertainty has

been suggested by Jamali¹³, and more recently by Modarres and Agarwal¹⁴. However, it can be seen that to implement this would require modifications to many of the commonly used codes.

When categorizing rather than ranking SSCs, the issue is somewhat less important, especially if the categorization is done in a conservative manner, and in an integrated fashion that would account for factors such as defense in depth and safety margins.

7.2 Modeling uncertainties

There are uncertainties that are not addressed explicitly in PRA models. Typical are uncertainties associated with how the model itself should be structured. For example, there might be an uncertainty related to the correct success criterion. Typically, such uncertainties are dealt with by making an assumption that the model chosen is the correct one. The user should be aware of such uncertainties and assess whether the choice of alternate modeling assumptions might significantly alter the conclusions of the analysis.

8 CONCLUSIONS

In order to address the issue of consistency with the NRC's

safety principles and the acceptance guidelines as presented in DG-1061³, it is important that some rational relationship be established between the criteria used to assess safety significance using importance measures such as the Risk Achievement Worth, and the acceptance criteria. It is more straightforward to establish criteria for assessing risk significance, using the Fussell–Vesely importance measure. In particular, since the applications may affect a large number of SSCs, and it is their integrated effect that is of interest, any criteria based on importance measures must reflect this. Clearly, the values chosen must reflect both the size and the constituency of the group. Because different applications address different groups of SSCs, it is difficult to establish a simple universal criterion at either the group or the individual basic event level that would not be extremely conservative for the majority of applications. However, it might be possible to establish a process by which appropriate criteria could be established as a first step in the categorization of SSCs. This would need to be followed up with further analysis using a mixture of qualitative arguments, based on an understanding of the role of the component groups, and well chosen sensitivity analyses to model the maximum reasonably expected change to parameters to further refine the categorization. An alternative to this top-down approach which focuses on the risk-significant SSCs is a bottom-up approach in which the set of non-risk-significant SSCs is constructed from the bottom of the list of SSCs ranked according to their RAW values. These alternatives would have to be tested with real PRA models to see if the approaches are practical, and give meaningful results.

The uncertainty in the probabilities of the basic events makes it difficult to determine a robust ranking of SSCs with respect to safety significance or risk significance. For those applications that require categorization of SSCs, the uncertainty plays a much less important role. However, consideration of uncertainty should be a factor in the integrated decision making process, particularly when considering the disposition of those events for which the uncertainty is considerably larger than that typical of the population of basic events.

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APPENDIX A RELATIONSHIPS BETWEEN IMPORTANCE MEASURES FOR GROUPS AND INDIVIDUAL MEMBERS OF GROUPS

Suppose the cutset solution for a simple system is as follows:

$$S = ABC + DEF,$$

then

$$RAW(C) = \frac{AB + DEF}{ABC + DEF}$$

$$RRW(C) = \frac{ABC + DEF}{DEF}$$

and

$$FV(C) = \frac{ABC}{ABC + DEF}$$

Suppose now $S = AB(C_1 + C_2) + DEF$, where $P(C) = P(C_1 + C_2)$, then

$$RAW(C_1) = \frac{ABC_2 + AB + DEF}{ABC + DEF} > RAW(C)$$

$$RAW(C_2) = \frac{ABC_1 + AB + DEF}{ABC + DEF} > RAW(C)$$

and

$$\text{RRW}(C_1) = \frac{ABC + DEF}{DEF + ABC_2} < \text{RRW}(C)$$

$$\text{RRW}(C_2) = \frac{ABC + DEF}{DEF + ABC_1} < \text{RRW}(C)$$

and

$$\text{FV}(C_1) = \frac{ABC_1}{AB(C_1 + C_2) + DEF}$$

$$\text{FV}(C_2) = \frac{ABC_2}{AB(C_1 + C_2) + DEF}$$

Note that $\text{FV}(C_1) + \text{FV}(C_2) = \text{FV}(C)$.

Thus, for the special case of a group of components in series, the Fussell–Vesely importance measure is additive.

Suppose now that $S = AB(C_1 \cdot C_2) + DEF$ where $P(C) = P(C_1 \cdot C_2)$. Then

$$\text{RAW}(C_1) = \frac{ABC_2 + DEF}{ABC + DEF} < \text{RAW}(C)$$

$$\text{RAW}(C_2) = \frac{ABC_1 + DEF}{ABC + DEF} < \text{RAW}(C)$$

and

$$\text{RRW}(C_1) = \frac{ABC + DEF}{DEF} = \text{RRW}(C) = \text{RRW}(C_2)$$

and

$$\text{FV}(C_1) = \frac{ABC_1 C_2}{ABC_1 C_2 + DEF} = \text{FV}(C_2)$$

While this is a simple relationship, for the more general case of a function whose cutset equation looked like $S = (A + H + C_1)(B + I + C_2)$, the FV measures look like:

$$\text{FV}(C_1) = \frac{C_1(B + I + C_2)}{(A + H + C_1)(B + I + C_2)}$$

$$\text{FV}(C_2) = \frac{C_2(A + H + C_1)}{(A + H + C_1)(B + I + C_2)}$$

Thus there is no simple relationship between the Fussell–Vesely measures of the group and its elements.

The examples given above are of simple system models. For a more general configuration, some of the group of components affected by the change may be in parallel and some in series, which would make the analysis more complicated, and the relationship of the individual importance measures to that of the group more complex.